

Setup

September 21st, 2026

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In this recitation class, we are going to install and setup some software on your machine so that it is properly configured for any future class.

1. Shell program

All commands listed in all recitation classes' tutorials or any other document have been validated on the following shell programs:

- bash (which is available out-of-the-box on Linux and MacOS):
<https://www.gnu.org/software/bash>
- Windows Subsystem for Linux (WSL): <https://wsl.dev>

If your operating system is Windows, please install [Windows Subsystem for Linux \(WSL\)](#).

2. Java

All recitation classes' tutorials have been validated under **Java 11**, thus it is important to make sure that this version is properly installed on your machine.

- Get Java SE Development Kit 11.0.27 from [here](#).
- Extract the downloaded file.
- Augment your environment variables with the full path to the `<extracted directory>`.

On Windows

In Search, search for and then select: System (Control Panel)

Click the Advanced system settings link.

Click Environment Variables. In the section System Variables find the `PATH` environment variable and select it. Click Edit.

In the Edit System Variable window, add `<path to the extracted directory>\bin` to the `PATH` environment variable. Click OK. Close all remaining windows by clicking OK.

On Linux

Run `export PATH="<path to the extracted directory>/bin:$PATH"` on the command line. (You might have to run the `export` every time you restart the computer. For a more permanent solution, please consider adding the export command to the `.bash_profile` file located in your home directory.)

On MacOS

Run `export PATH="<path to the extracted directory>/Contents/Home/bin:$PATH"` on the command line. (You might have to run the `export` every time you restart the computer. For a more permanent solution, please consider adding the export command to the `.bash_profile` file located in your home directory.)

- Open a command line and run the following command to check whether Java is correctly installed:

Shell

```
$ java -version
```

You should see something like this

None

```
java version "11.0.27" 2025-04-15 LTS
Java(TM) SE Runtime Environment 18.9 (build 11.0.27+8-LTS-232)
Java HotSpot(TM) 64-Bit Server VM 18.9 (build 11.0.27+8-LTS-232,
mixed mode)
```

3. Apache Maven

[Apache Maven](#) is a software project management and comprehension tool. Based on the concept of a project object model (POM), Maven can manage a project's build, reporting and documentation from a central piece of information.

- Get the latest version of [Apache Maven](#) from [here](#).
- Extract the downloaded file.
- Augment your environment variables with the full path to the `<extracted directory>/bin` (as you did for Java).
- Open a command line and run the following command to check whether Maven is correctly installed:

None

```
$ mvn -v
```

You should see something like this

None

Apache Maven 3.9.11 (3e54c93a704957b63ee3494413a2b544fd3d825b)

4. IntelliJ IDEA Community Edition

Get the latest version of IntelliJ IDEA Community Edition and install it:

Windows (x64): available [here](#).

Windows (ARM): available [here](#).

MacOS (x64): available [here](#).

MacOS (ARM): available [here](#).

Linux (Debian, Ubuntu, and any other Debian-based distributions):

None

```
$ sudo apt install snap
$ sudo snap install intellij-idea-community
--channel=2025.2.2/stable --classic
```

5. Demonstration project, a Pacman like game

- Get the Java demo project from [here](#).
- Extract the downloaded file.
- Open it on IntelliJ.
- Set the Java version (in case it has not been automatically set by the operating system):
File -> Project Structure
Point the IDE to the Java version you previously installed in Section 1, i.e., in the SDK box, select the 'Add JDK from disk...' and navigate to your Java installation.
Finally, in the 'Language level:' select 'SDK default'.
Click Apply and then OK.
- Set the Maven version (in case it has not been automatically set by the operating system):
Open IntelliJ Settings
Navigate to Build, Execution, Deployment -> Build Tools -> Maven
Point the IDE to the Maven version you previously installed in Section 2, i.e., in the Maven home path box, select the path to your Maven installation and then Apply and OK.
- Open the `pt.up.fe.demo.Main` class and run it.
- Enjoy it. :-)

Done!